

Signed-Pipeline vs Live — FRESH Re-validation (2026-07-09)

Revision: 1 Last modified: 2026-07-09T17:15:00Z

Fresh, single-owner (§11.4.119) re-run of the signed OTA pipeline against a **freshly booted REAL system** (real ota-server + real PostgreSQL, rootless podman). Supersedes the 2026-06-23 UNTRACKED/stale 15/15 evidence per §11.4.108 (no stale evidence — everything below was re-run today from a clean boot).

Host: the-factory (user milos), podman 5.7.1 rootless, Go 1.26.4, x86_64. Boot target: milos@localhost (the harness's default thinker.local was unreachable — DNS Name or service not known; the run was pointed at the local rootless-podman host over ssh-to-localhost, which the harness supports).

1. Boot the real system + /readyz + admin login — PASS

Command:

```
TARGET=milos@localhost REMOTE_USER=milos bash tests/lib/
boot_real_system.sh --up
```

Result (boot_standalone.log): cross-compiled a static linux/amd64 ota-server (22,618,274 bytes), podman-compose build + ordered up (postgres-first, waited for pg_isready, then ota-server), /readyz -> 200 {"status": "ready"}. BASE_URL=http://127.0.0.1:18080.

Independent capture (boot_readyz_login.txt), against the live loopback: - GET /readyz -> **HTTP 200** {"status": "ready"} - GET /healthz -> **HTTP 200** {"status": "ok"} - POST /api/v1/auth/login (admin@helix.system) -> **HTTP 200**, real DB-backed access_token issued (redacted), roles: ["admin", "operator", "viewer"]. - Containers: helix-ota-system_postgres_1 Up (healthy) :55480->5432, helix-ota-system_ota-server_1 Up (healthy) :18080->8080.

Standalone stack then torn down (bash boot_real_system.sh --down) — orphan check returned NONE (§11.4.14).

2. tests/e2e/pipeline_signed_live.sh — FRESH 15/15 PASS

Command:

```
TARGET=milos@localhost REMOTE_USER=milos \  
EVIDENCE_DIR=docs/qa/20260709-signed-pipeline-revalidate \  
bash tests/e2e/pipeline_signed_live.sh
```

Self-booted the real system in CALLER-PUBKEY MODE (live server trusts the caller's ephemeral ed25519 pubkey; real Postgres), /readyz -> 200 (run.log), drove the full pipeline, self-tore-down. **PASS=15 FAIL=0 SKIP=0, exit=0** (SUMMARY.txt). Per-step captured evidence (real HTTP responses, tokens redacted):

Step	Assertion	HTTP	Evidence file
01	live admin login, access_token	200	step01_login.txt
02	anti-bluff: bogus signature REJECTED (SIGNATURE_INVALID)	422	step02_antibluff_bogus_sig_reject.txt
03	upload BASE v1.0.0 (caller-signed) verified=true	201	step03_upload_base_signed.txt
04	upload TARGET v1.1.0 (caller-signed) verified=true	201	step04_upload_target_signed.txt
05	GET target artifact persisted (id echoes)	200	step05_get_target_artifact.txt
06	create BASE release	201	step06_release_base.txt
07	create TARGET release	201	step07_release_target.txt
08	register device on 1.0.0 (device_token issued)	201	step08_register_device.txt
09	deploy TARGET release	201	step09_deploy_target.txt
10	rollout create (echoes deployment_id)	201	step10_rollout_create.txt

Step	Assertion	HTTP	Evidence file
11	rollout get	200	step11_rollout_get.txt
12	rollout evaluate -> action=advance	200	step12_rollout_evaluate.txt
13	register base->target delta	201	step13_register_delta.txt
14	PAYOFF: device on 1.0.0 polls /client/ update, RECEIVES signed 1.1.0 (version, release, sha256 echoes signed TARGET, signature present, delta base_version=1.0.0 sha matches)	200	step14_device_poll_receives_signed_update.txt
15	control: device already on 1.1.0 gets no update	204	step15_device_on_target_poll.txt

Step-14 captured body (real): version:"1.1.0", sha256:91ae83856bdd..., signature:"0xSx/DrERKBN..VBg==", delta.base_version:"1.0.0", delta.sha256:7dfb18b3336a... — the device receives the correct SIGNED update through the full real-DB pipeline.

Post-run orphan check: NONE_CLEAN (§11.4.14 quiescence — no leftover containers).

3. Challenges bank dry-run — PASS (20/0/0)

Command:

```
bash tools/helixqa/run_bank.sh --dry-run
```

Result (challenges_dryrun.log): **20 passed / 0 failed / 0 skipped (mode=dry), RESULT: PASS**, exit=0. Honest boundary (§11.4.6): --dry-run is a STATIC coverage audit (bank structure + declared evidence_artifact existence) — it touches nothing live and is NOT a live challenge execution. A live full-bank run requires HELIX_ADMIN_PASSWORD and each challenge self-boots/tears-down.

4. Teardown / cleanup — PASS

Both the standalone boot and the pipeline's self-teardown removed their project-scoped stacks; post-run `podman ps -a | grep helix-ota-system` returned none. No source files modified (`git status`: only this new evidence dir added).

SKIP-with-reason

None. The default remote host `thinker.local` was unreachable, but the run was NOT skipped — it was pointed at the local rootless-podman host, booted, and proven live. No fabricated PASS; no metadata-only PASS.

Evidence files (this run)

- `boot_standalone.log`, `boot_standalone.stdout` — standalone boot
- `boot_readyz_login.txt` — independent `/readyz` + `/healthz` + admin login capture
- `run.log` — `pipeline_signed_live` full run log (self-boot readyz)
- `step01..step15 *.txt` — per-step captured HTTP responses (redacted)
- `SUMMARY.txt` — `PASS=15 FAIL=0 SKIP=0 exit=0`
- `pipeline_console.log` — full console of the pipeline run
- `challenges_dryrun.log` — challenges bank dry-run (20/0/0 PASS)